



Basic Principles of Medicine 1

Module: Musculoskeletal System | IMCQ3

IMCQ 3

*School of Medicine,
St. George's University*

1. A 65-year-old man is brought to the emergency department because of difficulty breathing. He is accompanied by his son who states that his father fell down the stairs 2 hours ago. His pulse is 100/min, respirations are 20/min, blood pressure is 145/83 mm Hg, and temperature is 36.5°C (97.7°F). Pulse oximetry on room air shows an oxygen saturation of 95%. Physical examination shows bruising on the anterior lower chest wall with notable depression of the area when the patient breathes in. The lung fields are clear on auscultation. Which of the following best describes the sign seen in the affected area during inspiration?

- A. Intercostal retraction
- B. Paradoxical breathing
- C. Bucket handle movement
- D. Pump handle movement
- E. Pectus excavatum

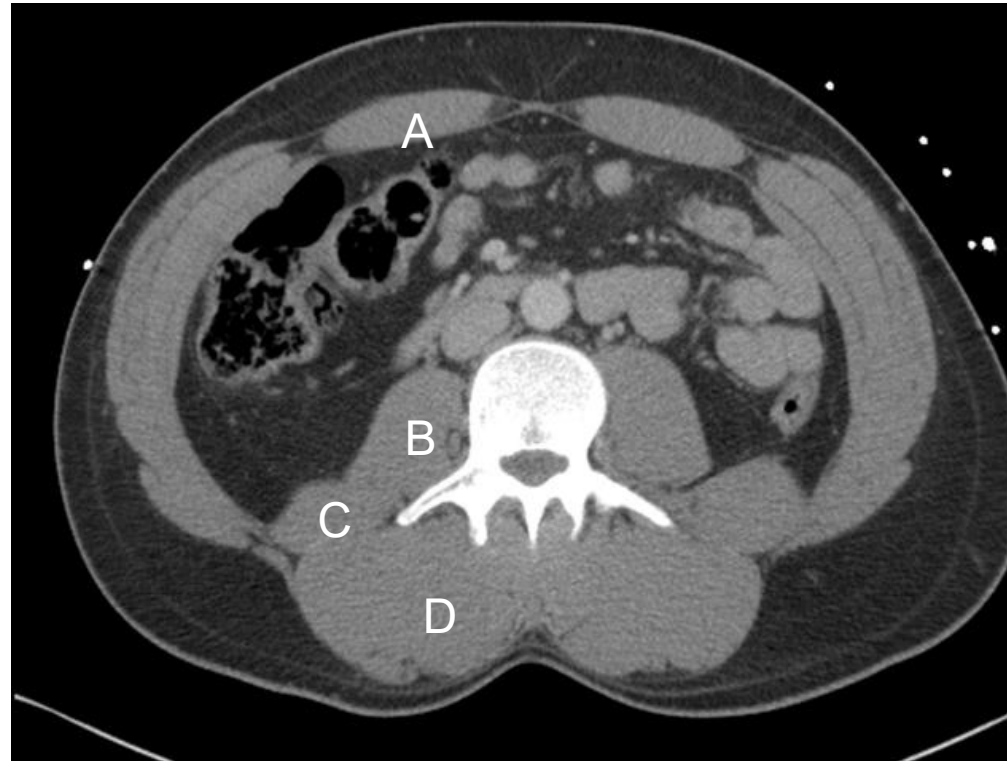
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SOM.MK.II.BPM1.1.MSK.1.ANAT.1217 Discuss and describe the the bucket-handle and pump-handle movements of the ribs during respiration.

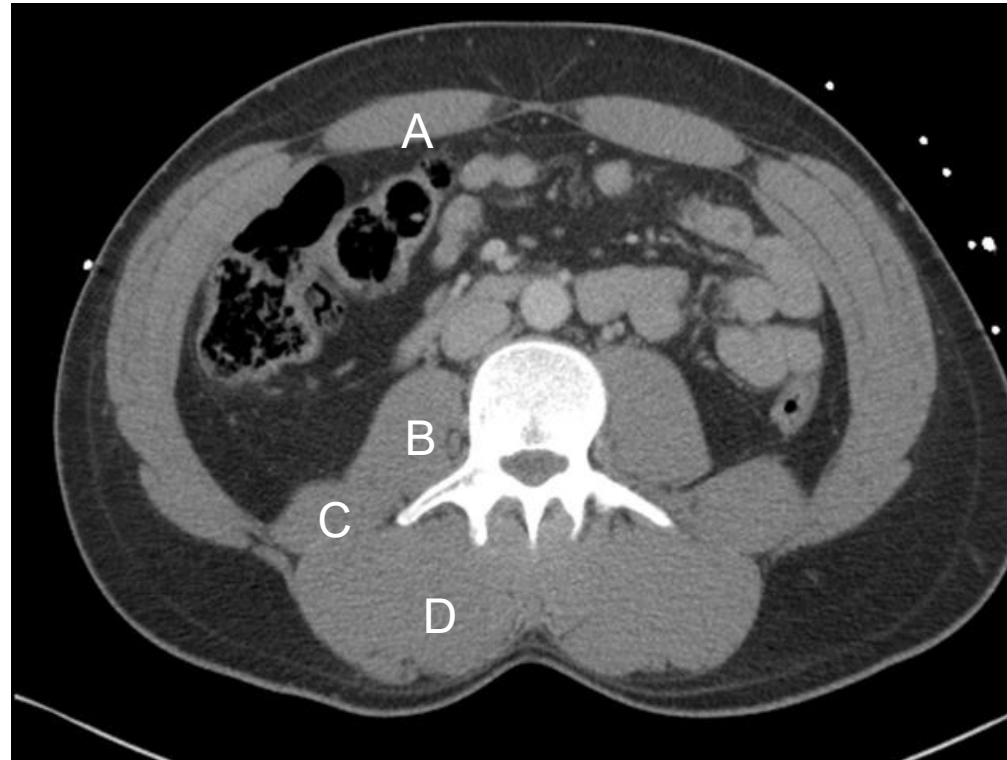
2. A 32-year-old man comes to the emergency department because of a sudden onset of rapid heartbeats that began 30 minutes ago. He describes the sensation as “fluttering” in his chest. His blood pressure is 100/70 mm Hg, pulse is 160/min and regular, respirations are 18/min, and temperature is 37°C (98.6°F). The physician instructs the patient to perform the Valsalva maneuver to relieve his symptoms. A CT image is shown. Which of the following structures indicated on the image is most important in performing the Valsalva maneuver?

- A. A
- B. B
- C. C
- D. D



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- B. B
- C. C
- D. D



SOM.MK.II.BPM1.1.MSK.1.ANAT.1248 List the general attachment, blood supply, innervations, and function of the muscles of the anterolateral abdominal wall.



3. A 2-year-old boy is brought to the physician because of a 2-month history of intermittent swelling in the left groin. There is no history of recent trauma. Physical examination shows both testes are present in the scrotum. A soft non-tender mass is palpated in the groin. The physician suspects an indirect inguinal hernia. Which of the following best describes the character of this mass?

- A. Enters the inguinal canal via the deep ring
- B. Enters the inguinal canal via the superficial ring
- C. Enters the inguinal canal medial to the inferior epigastric vessels
- D. Occurs above the arcuate line
- E. Enters inguinal canal via obturator foramen

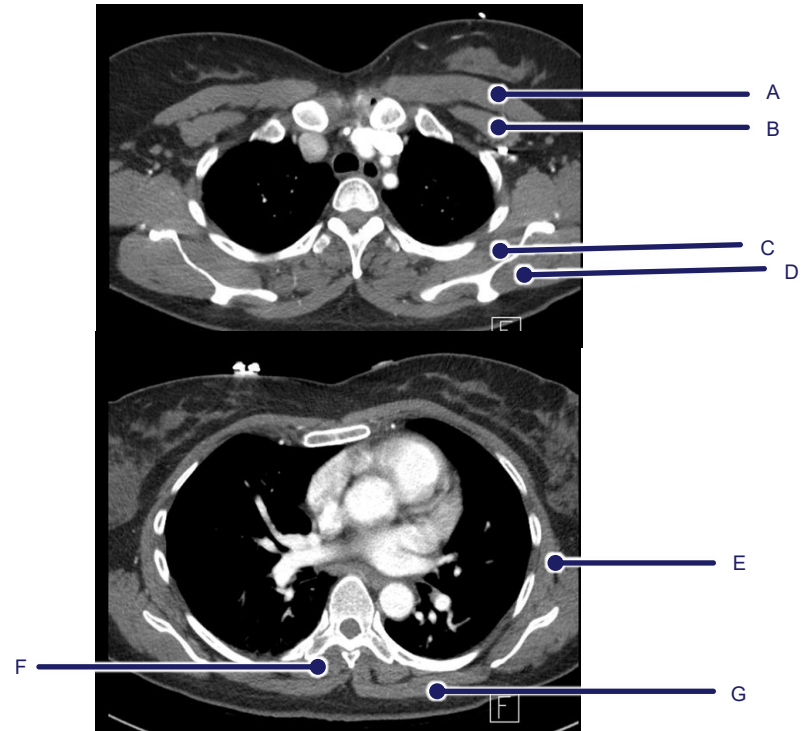
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SOM.MKII.BPM1.1.MSK.1.ANAT.1253 Explain the difference between a direct and an indirect inguinal hernia.

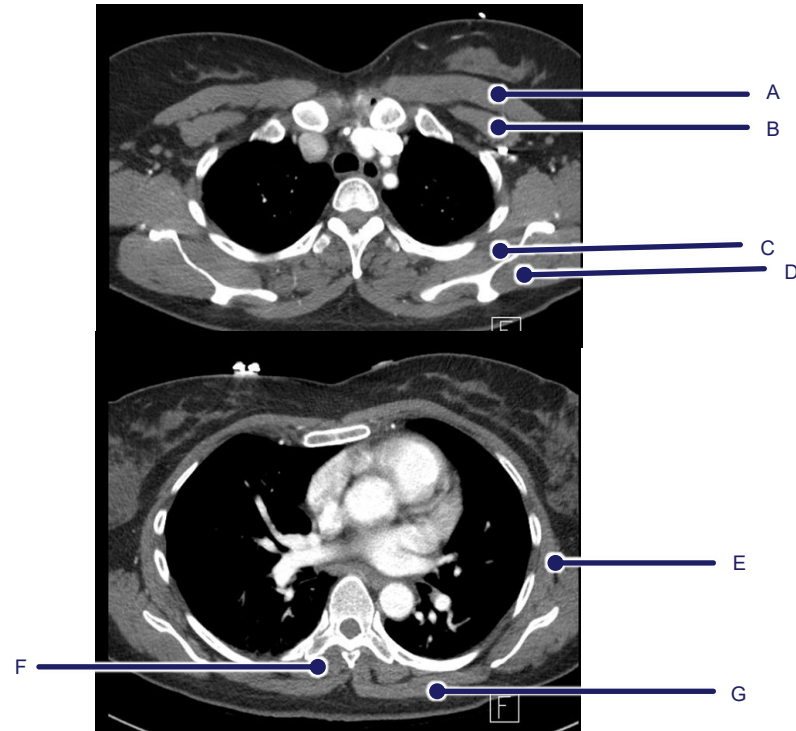
4. A 42-year-old woman comes to the physician because of a 1-week history of difficulty moving her left limb. She underwent radical mastectomy 1 week ago. Physical examination shows weakened lateral rotation of the shoulder joint. The nerve supplying which of the following labeled muscles is most likely damaged?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F
- G. G



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- C. C
- D. **D**
- E. E
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- G. G



SOM.MKII.BPM1.1.MSK.1.ANAT.1126 List the general attachments, innervations and actions of the muscles that contribute to the walls of the axilla.

5. A 26-year-old primigravida woman comes to the physician because of a 2-month history of tender swelling just below her left armpit. She first noticed a small, mole-like area in this region during adolescence, but it had never caused discomfort before. The swelling has enlarged during the last 2 months of pregnancy. Physical examination shows a 2-cm, soft, movable, non-erythematous nodule in the left axilla, distinct from the breast tissue. Which of the following best explains the embryologic origin of this finding?

- A. Persistence of the second branchial cleft
- B. Incomplete canalization of the lactiferous duct
- C. Duplication of the apocrine sweat gland
- D. Hyperplasia of sebaceous glands in the axilla
- E. Failure of regression of the mammary ridge along the milk line

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SOM.MK.II.BPM1.1.MSK.1.ANAT.1228 Describe polythelia, polymastia, amastia, gynecomastia and their clinical significance.

6. A 47-year-old man comes to the physician because of a 3-week history of scrotal pain. Physical examination shows an absent cremasteric muscle reflex. Which of the following is the most likely structure from which the muscle was derived?

- A. Internal abdominal oblique muscle and aponeurosis
- B. External abdominal oblique muscle aponeurosis
- C. External abdominal oblique muscle
- D. Transversalis fascia
- E. Parietal peritoneum

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7. A 51-year-old man is admitted to the hospital with severe dyspnea. An x-ray of the chest shows a tension pneumothorax. Adequate local anesthesia of the chest wall prior to the insertion of a chest tube is administered. Which of the following layers is most likely the deepest that must be infiltrated with the local anesthetic to achieve adequate anesthesia?

- A. Endothoracic fascia
- B. Intercostal muscles
- C. Parietal pleura
- D. Subcutaneous fat
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SOM.MK..BPM1.1.MSK.1.ANAT.1240 Describe thoracocentesis and explain why the needle is passed in the lower border of the intercostal space

8. A 52-year-old woman comes to the physician because of a 6-month history of a lump in her left breast. Physical examination shows pitting over the breast in the area of the lump (peau d'orange). Mammography shows a 6-cm mass in the lower lateral quadrant. Which of the following structures is most likely affected, leading to the changes seen in the patient's skin?

- A. Branches of cutaneous pectoral nerves
- B. Lactiferous ducts
- C. Mammary arteries and veins
- D. Mammary lymphatics
- E. Suspensory (cooper) ligaments

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SOM.MKII.BPM1.1.MSK.1.ANAT.1229 Explain the clinical importance of the lymphatic drainage of a cancerous mammary gland.

9. A 42-year-old man comes to the physician because of a 2-week history of difficulty flexing his hip, sometimes accompanied by abdominal pain when he attempts to move the hip. He recently began having more frequent dance rehearsals for an upcoming event. Injury of a hip flexor that stabilizes the spine and pelvis is suspected. The injured muscle is indicated by the tip of the arrow in the shown image. Which of the following nerve roots most likely provides innervation to the injured muscle?

- A. Ventral rami of L1-3
- B. Dorsal rami of L1-3
- C. Ventral rami of L2-4
- D. Dorsal rami of L2-4
- E. Ventral rami of L3-5
- F. Dorsal rami of L3-5



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- D. Dorsal rami of L2-4
- E. Ventral rami of L3-5
- F. Dorsal rami of L3-5





10. A 45-year-old man comes to the physician because of a 4-day history of painful swelling in his groin. He says that the swelling suddenly appeared three weeks ago after he lifted a heavy box from the floor but recently became painful. During surgical repair, the surgeon identifies a vessel located lateral to the herniated bowel. Which of the following features best describes this vessel?

- A. Provides blood supply to the testes
- B. Medial border of Hesselbachs' triangle
- C. Medial to the deep inguinal ring
- D. Forms the medial umbilical fold
- E. Remnant of urachus

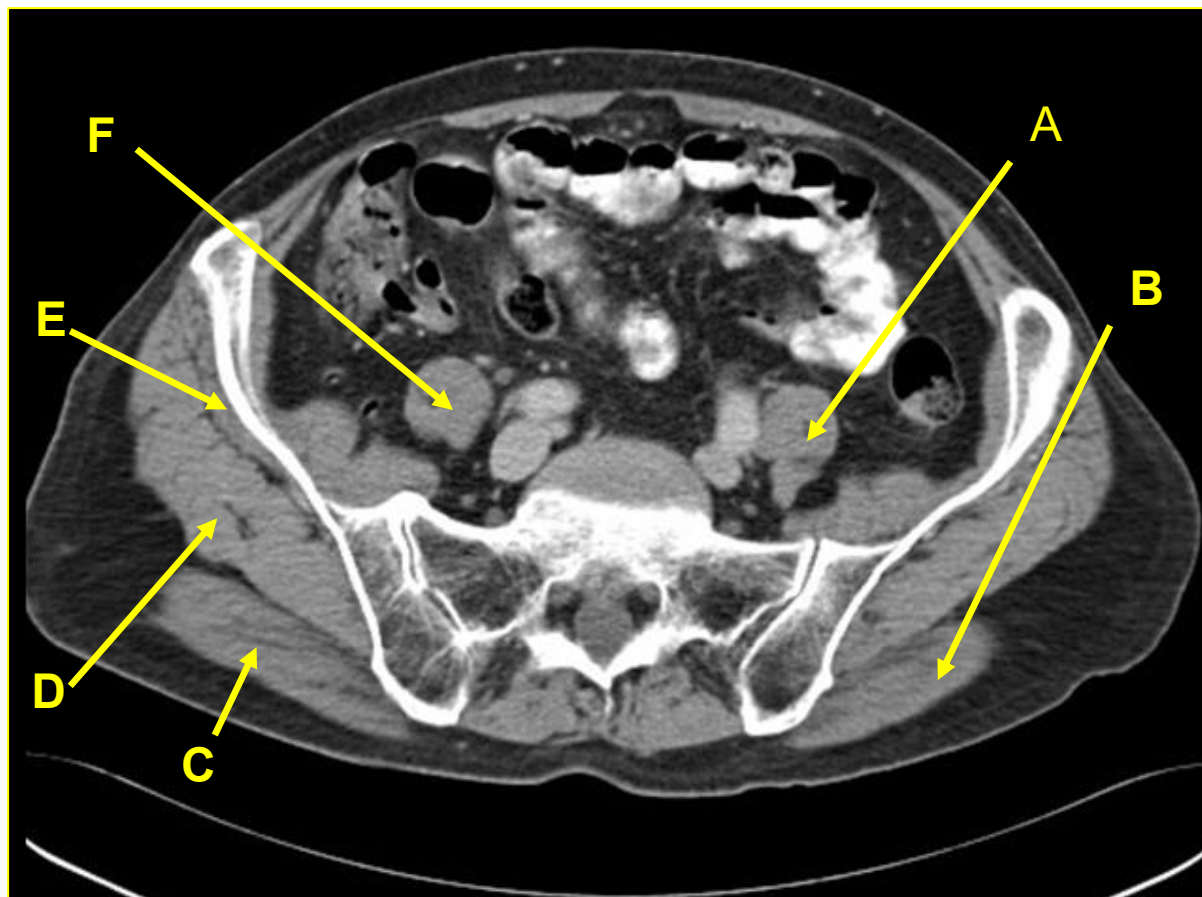
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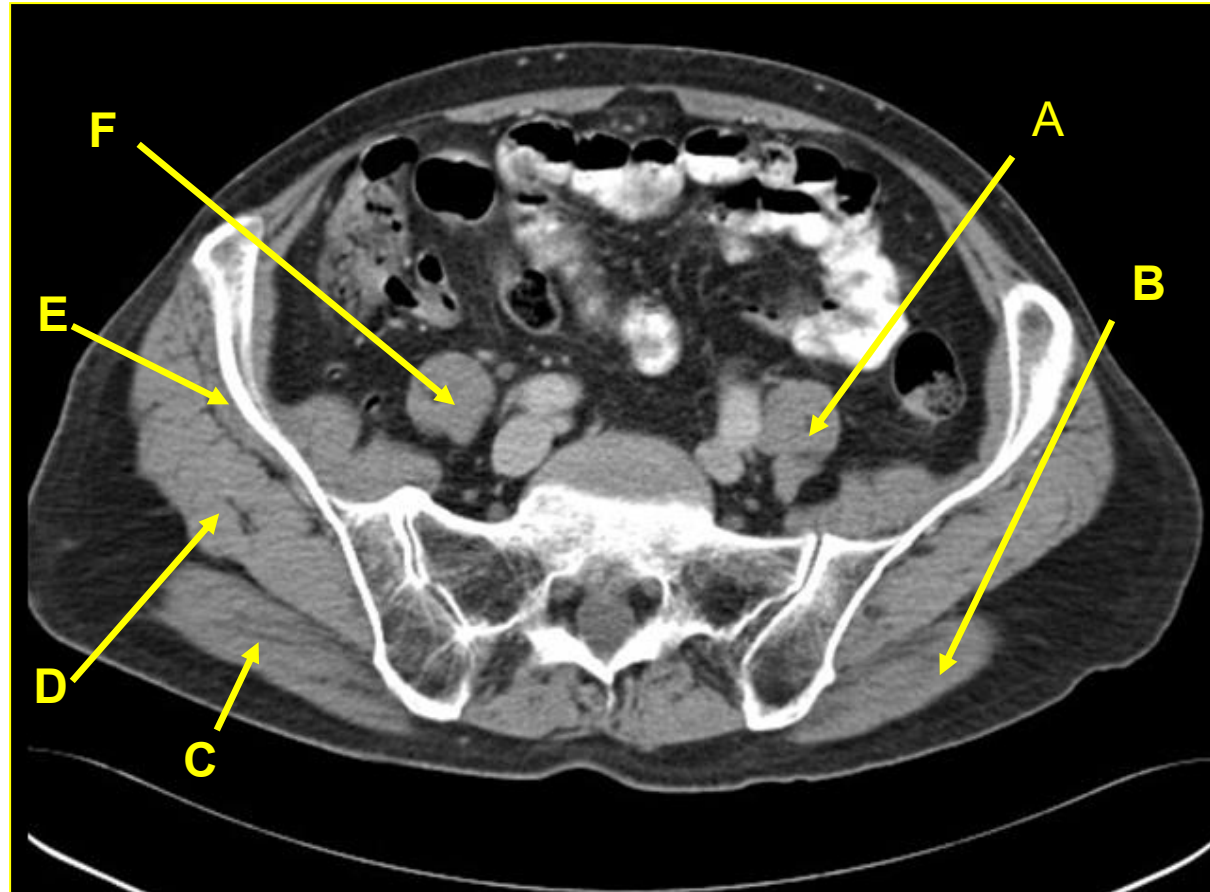
11. A 27-year-old woman is brought to the emergency department because of a 4-day history of fever and severe back pain on her right side. Physical examination shows sharp, right-sided lumbar paraspinal pain that radiates down to her right anterior thigh. She resists flexion of her right hip joint. Which of the following labeled muscles is most likely responsible for this patient's findings?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F



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- A. A
- B. B
- C. C
- D. D
- E. E
- F. **F**



SOM.MK.II.BPM1.1.MSK.1.ANAT.1266 List the innervations, blood supply, general attachments and actions of the muscles of the posterior abdominal wall.

12. A 42-year-old man comes to the physician for a post-operative visit after being admitted to the hospital following a femoral hernia repair. The surgical notes indicate that the surgery was complicated by the unusually large size of the herniated intestine, necessitating an open approach instead of a laparoscopic repair. Physical examination shows decreased sensation in the skin of the upper anterior thigh (skin over the femoral triangle) and absent cremasteric reflex. Which of the following best describes the anatomical relation of the affected nerve to the muscle indicated by the arrow?

- A. Medial
- B. Lateral
- C. Anterior
- D. Posterior
- E. Inferior



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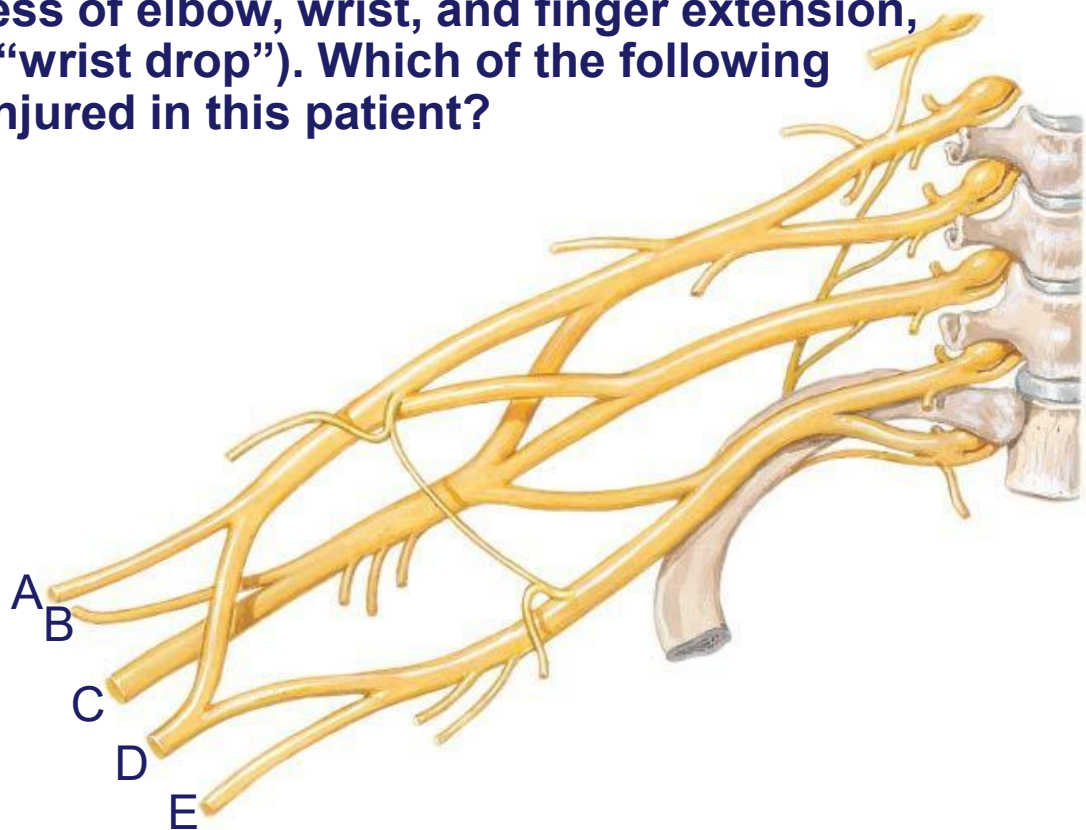
- A. Medial
- B. Lateral
- C. Anterior
- D. Posterior
- E. Inferior



SOM.MKII.BPM1.1.MSK.1.ANAT.1267 Describe the origin of the lumbar plexus and list its branches and their distributions.

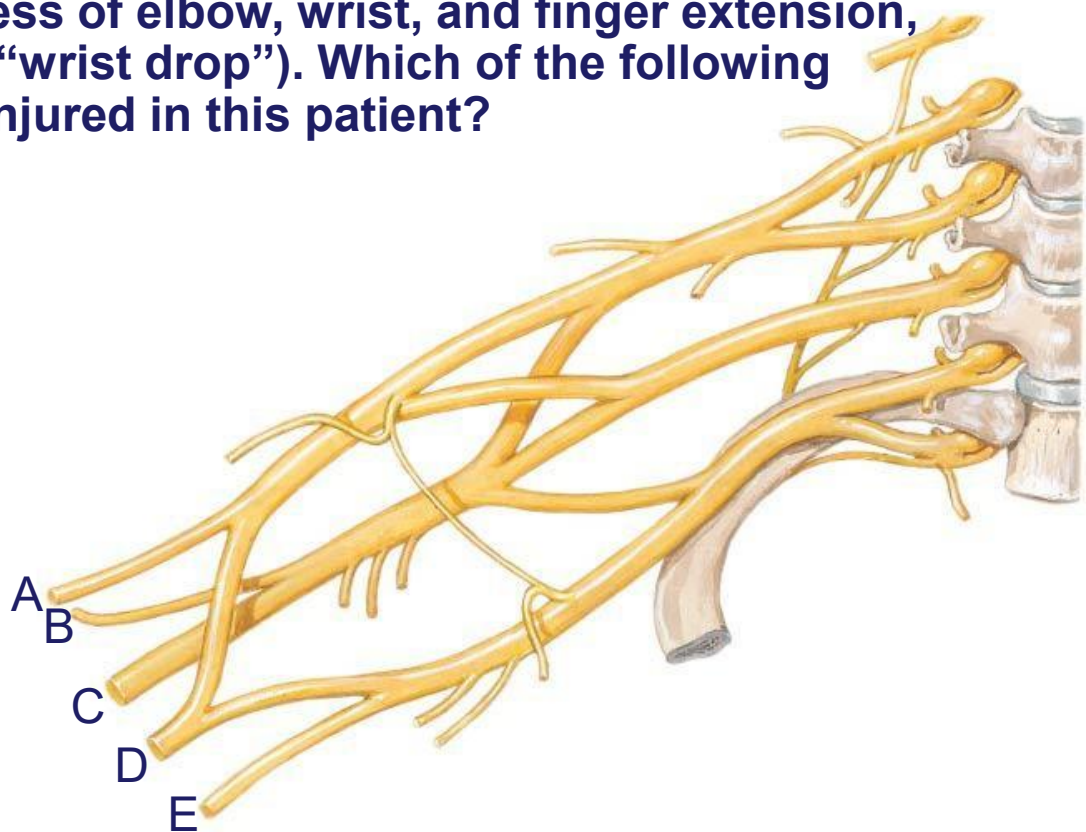
13. A 42-year-old man is brought to the emergency department because of a 5-hour history of upper limb weakness. He is unable to extend his right wrist and fingers. He fell asleep in a chair with his right arm hanging over the backrest after several hours of drinking alcohol. When he woke up, his hand was limp, and he could not straighten his wrist. Physical examination shows loss of sensation over the posterior arm, posterior forearm, dorsum of the hand, and the dorsal aspect of the lateral two and a half fingers. There is weakness of elbow, wrist, and finger extension, and the hand rests in flexion (“wrist drop”). Which of the following labeled nerves is most likely injured in this patient?

- A. A
- B. B
- C. C
- D. D
- E. E



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- A. A
- B. B
- C. **C**
- D. D
- E. E





14. A 3-year-old boy is brought to the emergency department because he refuses to move his right arm after a fall earlier in the day. His older sister, attempting to prevent him from hitting the ground, suddenly pulled upward on his arm. The boy immediately cried and has since kept the arm still. Physical examination shows the child holding his right arm flexed at the elbow and pronated against his abdomen. Any attempt to move the elbow elicits crying. There is no swelling, deformity, or bruising. Which of the following structures is most likely affected in this patient's condition?

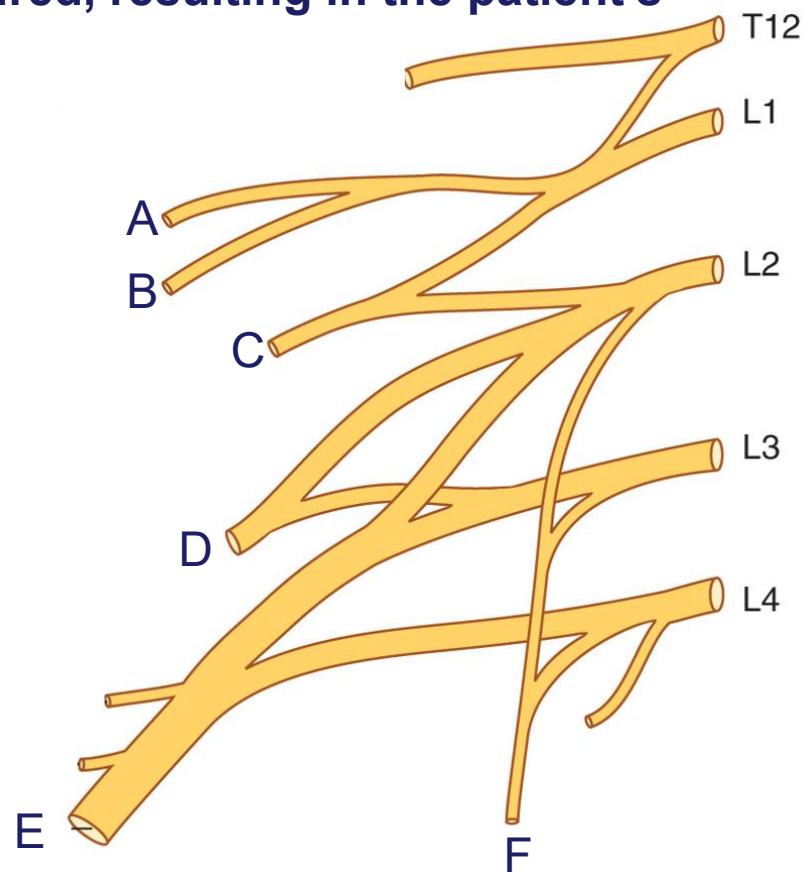
- A. Annular ligament
- B. Biceps tendon
- C. Interosseous membrane
- D. Radial collateral ligament
- E. Ulnar collateral ligament

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- C. Interosseous membrane
- D. Radial collateral ligament
- E. Ulnar collateral ligament

15. A 46-year-old woman comes to the physician because of a 3-month history of burning pain and tingling over the outer aspect of her right thigh. The discomfort is worse when she stands or walks for long periods and improves when she sits. She has recently gained 10 kg (22 lb). Height is 165 cm (5 ft 5 in) and weight is 85 kg (187 lb); BMI = 31 kg/m². Physical examination shows loss of sensation to light touch and pinprick over the anterolateral thigh, but no motor strength deficits. Which of the following labeled nerves (A–E) is most likely injured, resulting in the patient's findings?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F



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A. A

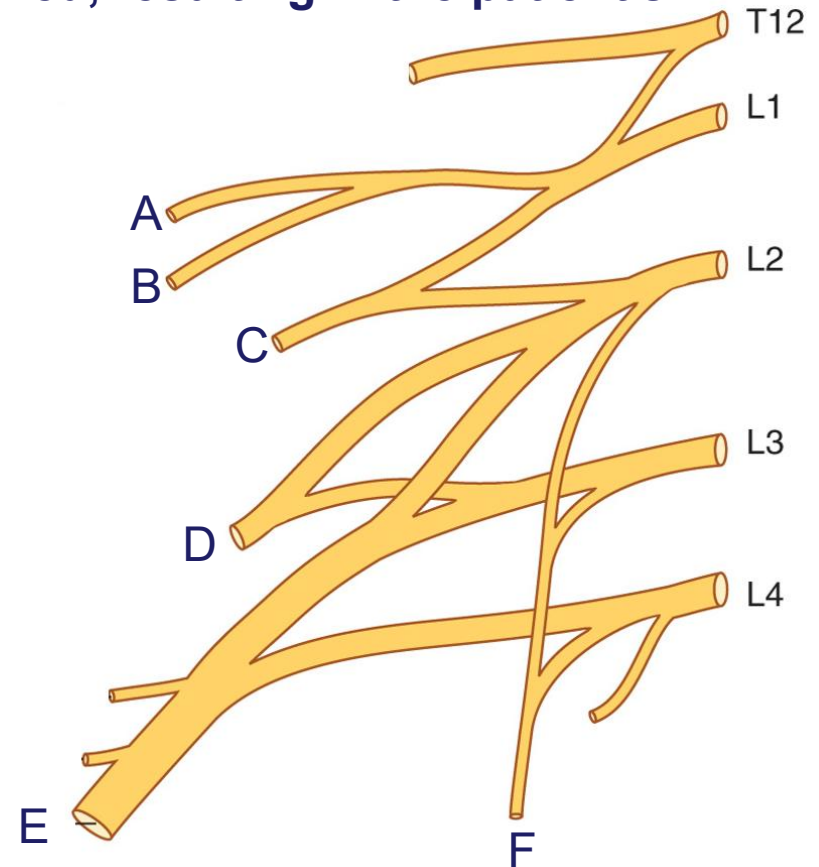
B. B

C. C

D. **D**

E. E

F. F



16. A 55-year-old woman comes to the physician because of a 3-month history of a painless lump in her right breast. Her mother died of breast cancer at age 60. Physical examination shows a firm, non-tender, immobile mass measuring 3 cm in diameter in the upper outer quadrant of the right breast. The overlying skin appears thickened with a slight dimpling but no erythema. A bloody discharge is noted from the nipple. Palpation reveals several firm, non-tender lymph nodes in the right axilla. A core needle biopsy confirms invasive ductal carcinoma. Which of the following lymph node groups is most likely the initial site of metastasis in this patient?

- A. Central axillary nodes
- B. Apical axillary nodes
- C. Anterior (pectoral) axillary nodes
- D. Internal thoracic (parasternal) nodes
- E. Subdiaphragmatic nodes

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SOM.MK.II.BPM1.1.MSK.1.ANAT.1232
Describe a sentinel lymph node and its clinical significance.